math-blurbs

self-adjoint

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This project is hosted on GitHub: https://github.com/self-adjoint/math-blurbs

0 Introduction

0.1 The QED Symbol

QED stands for *quod erat demonstrandum* (Latin for "that which was to be demonstrated"), and is used to signify the end of a mathematical proof. Most mathematicians use either a white square □ or a black square ■ for this. Paul J., Sally, Jr. (1933 – 2013) famously used a sketch of himself in an eyepatch[1] as the QED symbol in his books.

Theorem 0.1. For any group of people, there are at least two people with the same number of friends.

Proof. Suppose there are n people in the group. Each person can have any number of friends from 0 to $n-1$ (so there are n possible values). However, there cannot be both a person with 0 friends (friends with no one) and a person with $n-1$ friends (friends with everyone). Thus, there must be at least two people with the same number of friends. 😎

The QED symbol I have chosen to use is the Smiling Face with Sunglasses emoji[2]. This version has been used by WhatsApp since 2017.

Default minted options: autogobble, breaklines, mathescape

```python
from collections.abc import Iterator

# This is an example with math. The n\textsuperscript{th} Fibonacci number is (φ^n − (1−φ)^n)/√5, where
↪ φ = (1+√5)/2 is the golden ratio.
class Math:
    @staticmethod
    def fib(n: int) -> Iterator[int]:
        """Fibonacci series up to n."""
        a, b = 0, 1
        while a < n:
            yield a
            a, b = b, a + b

result = sum(Math.fib(42))
print(f"The answer is {result}")
```

title

some text

title

some text

title

some text

This is **NOT** the same. Oh, but **NOT**. This is **NOT** * the same. Oh, but **NOT** *.

[1] He was legally blind in his left eye, and nicknamed "The Math Pirate" and "Professor Pirate" by his students.

[2] Not to be confused with an emoticon, which refers to things like :) and :(and were mainly used before emojis were introduced into Unicode 6.0.

0.1

The exercise.

Solution

The solution.

```python
from collections.abc import Iterator


# This is an example
class Math:
    @staticmethod
    def fib(n: int) -> Iterator[int]:
        """Fibonacci series up to n."""
        a, b = 0, 1
        while a < n:
            yield a
            a, b = b, a + b


result = sum(Math.fib(42))
print(f"The answer is {result}")
```

Now another style:

```python
print('Did this work?')
```

Also, inline code looks like `this`.

$\begin{pmatrix} 6 & 5 \\ 4 & 3 \end{pmatrix}$

$\lim_{k\to\infty} f(k) = \lim\limits_{k\to\infty} f(k) = 0$

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